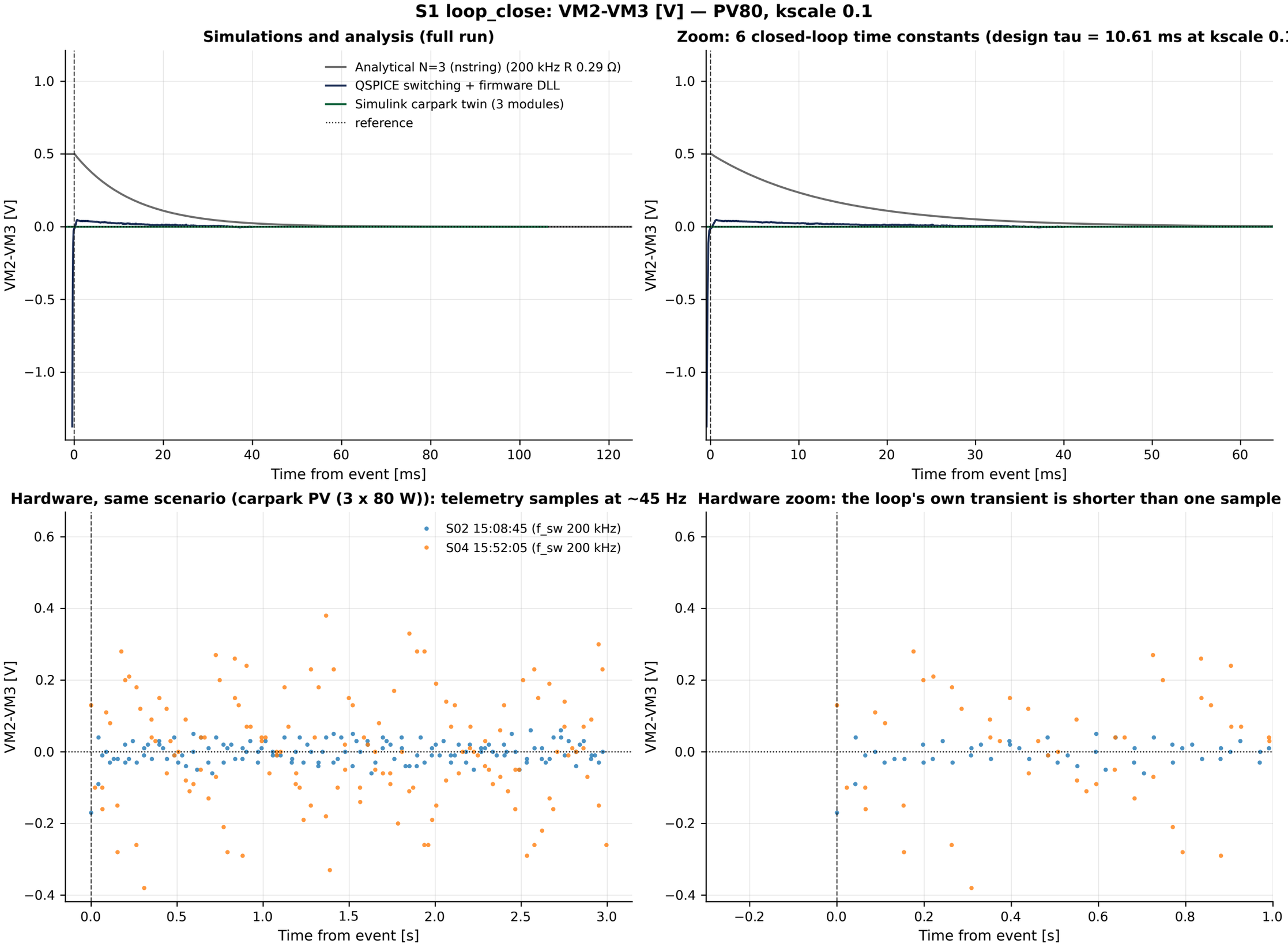
S1 loop_close: VM2-VM3 [V] — PV80, kscale 0.1
Simulations and analysis (full run)
Zoom: 6 closed-loop time constants (design tau = 10.61 ms at kscale 0.1
Analytical N=3 (nstring) (200 kHz R 0.29 Ω)
QSPICE switching + firmware DLL
Simulink carpark twin (3 modules)
reference
VM2-VM3 [V]
Time from event [ms]
Hardware, same scenario (carpark PV (3 x 80 W)): telemetry samples at ~45 Hz
Hardware zoom: the loop's own transient is shorter than one sample
S02 15:08:45 (f_sw 200 kHz)
S04 15:52:05 (f_sw 200 kHz)
Time from event [s]